

Sistema lineare

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Metodo iterativi

$$x^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \curvearrowright x^{(1)} \curvearrowright x^{(2)} \curvearrowright \dots \Big| \leadsto \underline{x}$$

Il metodo $\bar{x}$ converge $\Leftrightarrow \lim_{k \rightarrow \infty} x^{(k)} = \underline{x}$

dove $\underline{x}$ è la soluzione esatta.

$$\begin{cases} 2x_1 - x_2 + x_3 = 1 \\ 2x_1 + 2x_2 + 2x_3 = 1 \\ -x_1 - x_2 + 2x_3 = 1 \end{cases}$$

Metodo di Jacobi

$$x^{(0)} = \begin{pmatrix} x_1^{(0)} \\ x_2^{(0)} \\ x_3^{(0)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} x_1^{(1)} = \frac{1 + x_2^{(0)} - x_3^{(0)}}{2} = \frac{1 + 0 - 0}{2} = \frac{1}{2} \\ \dots \end{cases}$$

$$\begin{cases} x_2^{(1)} = \frac{1 - 2x_1^{(0)} - 2x_3^{(0)}}{2} = \frac{1 - 2 \cdot 0 - 2 \cdot 0}{2} = \frac{1}{2} \\ x_3^{(1)} = \frac{1 + x_1^{(0)} + x_2^{(0)}}{2} = \frac{1 + 0 + 0}{2} = \frac{1}{2} \end{cases}$$

$$X^{(1)} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$\begin{cases} x_1^{(2)} = \frac{1 + x_2^{(1)} - x_3^{(1)}}{2} = \frac{1 + \frac{1}{2} - \frac{1}{2}}{2} = \frac{1}{2} \\ x_2^{(2)} = \frac{1 - 2x_1^{(1)} - 2x_3^{(1)}}{2} = \frac{1 - 2 \cdot \frac{1}{2} - 2 \cdot \frac{1}{2}}{2} = -\frac{1}{2} \\ x_3^{(2)} = \frac{1 + x_1^{(1)} + x_2^{(1)}}{2} = \frac{1 + \frac{1}{2} + \frac{1}{2}}{2} = 1 \end{cases}$$

$$X^{(2)} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix}$$

$$\begin{cases} x_1^{(k+1)} = \frac{1 + x_2^{(k)} - x_3^{(k)}}{2} \\ x_2^{(k+1)} = \frac{1 - 2x_1^{(k)} - 2x_3^{(k)}}{2} \\ x_3^{(k+1)} = \frac{1 + x_1^{(k)} + x_2^{(k)}}{2} \end{cases}$$

Gauss - Seidel

(k)

Gauss - Seidel

$$\begin{cases} x_1^{(k+1)} = \frac{1 + x_2^{(k)} - x_3^{(k)}}{2} \\ x_2^{(k+1)} = \frac{1 - 2x_1^{(k+1)} - 2x_3^{(k)}}{2} \\ x_3^{(k+1)} = \frac{1 + x_1^{(k+1)} + x_2^{(k+1)}}{2} \end{cases}$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} x_1^{(1)} = \frac{1 + 0 - 0}{2} = \frac{1}{2} \\ x_2^{(1)} = \frac{1 - 2x_1^{(1)} - 2x_3^{(0)}}{2} = \frac{1 - 2 \cdot \frac{1}{2} - 2 \cdot 0}{2} = 0 \\ x_3^{(1)} = \frac{1 + x_1^{(1)} + x_2^{(1)}}{2} = \frac{1 + \frac{1}{2} + 0}{2} = \frac{3}{4} \end{cases}$$

Sistema generale.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{cases}$$

$$x_i = \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j - \sum_{j=i+1}^n a_{ij}x_j}{a_{ii}} \quad i=1 \dots n$$

$$x_i = \frac{b_i - \sum_{j=1, j \neq i}^n a_{ij} x_j}{a_{ii}} \quad (i=1 \dots n)$$

Jacobi

$$x_i^{(k+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^n a_{ij} x_j^{(k)}}{a_{ii}} \quad i=1 \dots n$$

Gauss-Seidel

$$x_i^{(k+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij} x_j^{(k)}}{a_{ii}} \quad i=1 \dots n$$

>> esercizioJacobiGaussSeidel

xJ =

1.0e+04 *

-1.8684

1.8684

-2.8025

Jacobi

kJ =

100

stimaerroreJ =

1.6270

ierJ =

1

→ non convergente!

xGs =

0.1668

Gauss Seidel

0.1668
-0.1668
0.5000

kGs =

16

stimaerroreGs =

8.2284e-04

ierGs =

0

>> x=A\b

x =

0.1667
-0.1667
0.5000

→ Convergente

soluzione di rifacimento

Soluzioni esatte

$$x = \begin{bmatrix} 0.1\bar{6} \\ -0.1\bar{6} \\ 0.5 \end{bmatrix}$$

calcolata
con
poco

>> xGs

xGs =

0.1668
-0.1668
0.5000

>> x-xGs

ans =

1.0e-03 *

-0.1043
0.1195
0.0076

→ STIMA

>> stimaerroreGs

stimaerroreGs =

8.2284e-04

>> norm(x-xGs)/norm(x)

ans =

→ l'errore è stato ben stimato

ans =

2.8728e-04

>>

→ ERRORE

stato ben s...